

scRNA+TCR-seq Reveals the Proportion and Characteristics of Dual TCR Treg Cells in Mouse Lymphoid and Non-lymphoid Tissues

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eLife Assessment

This study reanalyzed previously published scRNA-seq and TCR-seq data to examine the proportion and characteristics of dual-TCR-expressing Treg cells in mice, presenting some **useful** insights into TCR diversity and immune regulation. However, the evidence is **incomplete**, particularly with respect to data interpretation, statistical rigor, and the functionality of dual-TCR Treg cells. The study is potentially of interest to immunologists studying T-cell biology.

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Abstract

As an essential molecule for Treg cell function, TCR plays a critical role in various immune responses including cell development, immune tolerance, and self-antigen recognition. Current research has identified Treg cells in both lymphoid and non-lymphoid tissues as important for maintaining immune homeostasis and preventing autoimmune reactions. However, the TCR pairing types and CDR3 characteristics of these Treg cells are largely unknown. Through innovative analysis of scRNA+TCR-seq data, we discovered that 10.1% to 21.4% of dual TCR Treg cells exist in mouse lymphoid and non-lymphoid tissues. Dual TCR Treg cells in different tissues exhibit differential phenotypic and TCR repertoire characteristics. These findings provide key insights into the complex biological functions of Treg cell V(D)J recombination, tolerance selection, development, and migration.

Introduction

The discovery of CD4+CD25+ Treg cells in 1995, has since been linked to the occurrence and development of various human diseases. In addition to their strong regulatory role in self-reactivity and excessive inflammation, Treg cells play an important homeostatic role in the

maintenance of tissue homeostasis, muscle damage repair, hair follicle regeneration, fat metabolism, among other complex cellular activities[1][2]. The diverse negative regulatory functions of Treg cells suggest unique characteristics for their TCR diversity, specificity, and memory responses. However, the compositional characteristics of Treg cell TCR, their dependency on antigen epitopes, activation, and maintenance mechanisms, remain largely unelucidated. Some studies have garnered widespread attention, including: V(D)J recombination of nTreg cells TCRs, self-tolerance selection, and migratory settlement with or without lineage characterization; differences in antigen induction and activation mechanisms for iTreg versus conventional helper and effector T cells; diversity and plasticity of Treg cells in lymphoid and non-lymphoid tissues, such as TCR CDR3, and their correspondence to regulatory functions.

Since their initial report in 1988[3], dual TCR expressing lymphocytes have been widely supported by experimental evidence and play important roles in physiological and pathological processes such as autoimmune tolerance, transplant rejection reactions, and T cell tumors[4][5][6][7][8][9]. Due to technical limitations, comparing the proportion, characteristics, origin, and mechanisms of single TCR and dual TCR T cells in a single sample of more than a thousand T cells has been challenging for immunologists. However, scRNA-seq combined with scTCR-seq can mark over 5,000 T cells from a single research sample at once, obtaining the pairing types of each T cell's TCR beta chain and alpha chain, CDR3 sequence composition characteristics, and complete mRNA expression data. This allows for the accurate analysis of TCR CDR3 characteristics in the development, differentiation, maturation, and response/tolerance processes of each T cell, as well as the corresponding expression of regulatory and effector molecules. This provides unprecedented opportunities for in-depth analysis of dual receptor T cells[8][9][10][11].

First, in the Treg study conducted by Oliver T et al.[12], which involved analyzing single-cell RNA sequencing and T cell receptor (scRNA+TCR-seq) data of Foxp3⁺ Treg cells sorted from mouse blood, kidney, liver, LPL, and pancreas under physiological conditions, we conducted a detailed comparative analysis of the proportions of single TCR Treg and dual TCR Treg; the usage, diversity, clonality, and overlap of CDR3 VJ; and the homogeneity and heterogeneity of characteristic marker mRNA expression. Additionally, to further explore the differences in dual TCR Treg cells between lymphoid and non-lymphoid tissues in mice, we selected the scRNA+TCR-seq data shared by Annekathrin T et al.[13], which includes sequencing results of CD4⁺CD25⁺ Treg cells sorted from mouse spleen, inguinal lymph nodes (iLN), and mesenteric lymph nodes (mLN) under physiological conditions, as well as CD3⁺ T cells (including Treg and non-Treg) sorted from skin tissue, and performed comparative analysis again.

Results

1. A high proportion of dual TCR Treg cells in mouse lymphoid and non-lymphoid tissues

The proportion of dual TCR Treg cells in mouse lymphoid tissues: spleen=21.4%; iLN=21.4%, mLN=21.0% (**Figure 1D**); The proportion of dual TCR Treg cells in mouse non-lymphoid tissues: blood=11.9%; kidney=11.2%; liver=13.6%, LPL=10.1%; pancreas=13.2%, The proportion of dual TCR T cells in mouse skin tissue: 18.6% (**Figure 3D**). Subsequently, a detailed analysis of the pairing of dual TCR T cells in the two sets of data was conducted, revealing that the TCR pairing types of dual TCR Treg include A+B1+B2, B+A1+A2 (the highest proportion), A1+A2+B1+B2 (the lowest proportion). There are significant differences in TCR pairing types between different tissues, with A1+A2+B1+B2 significantly increased in skin and kidney, and B+A1+A2 significantly higher in spleen and liver than in other groups (**Figure 1D**, **Figure 3D**).

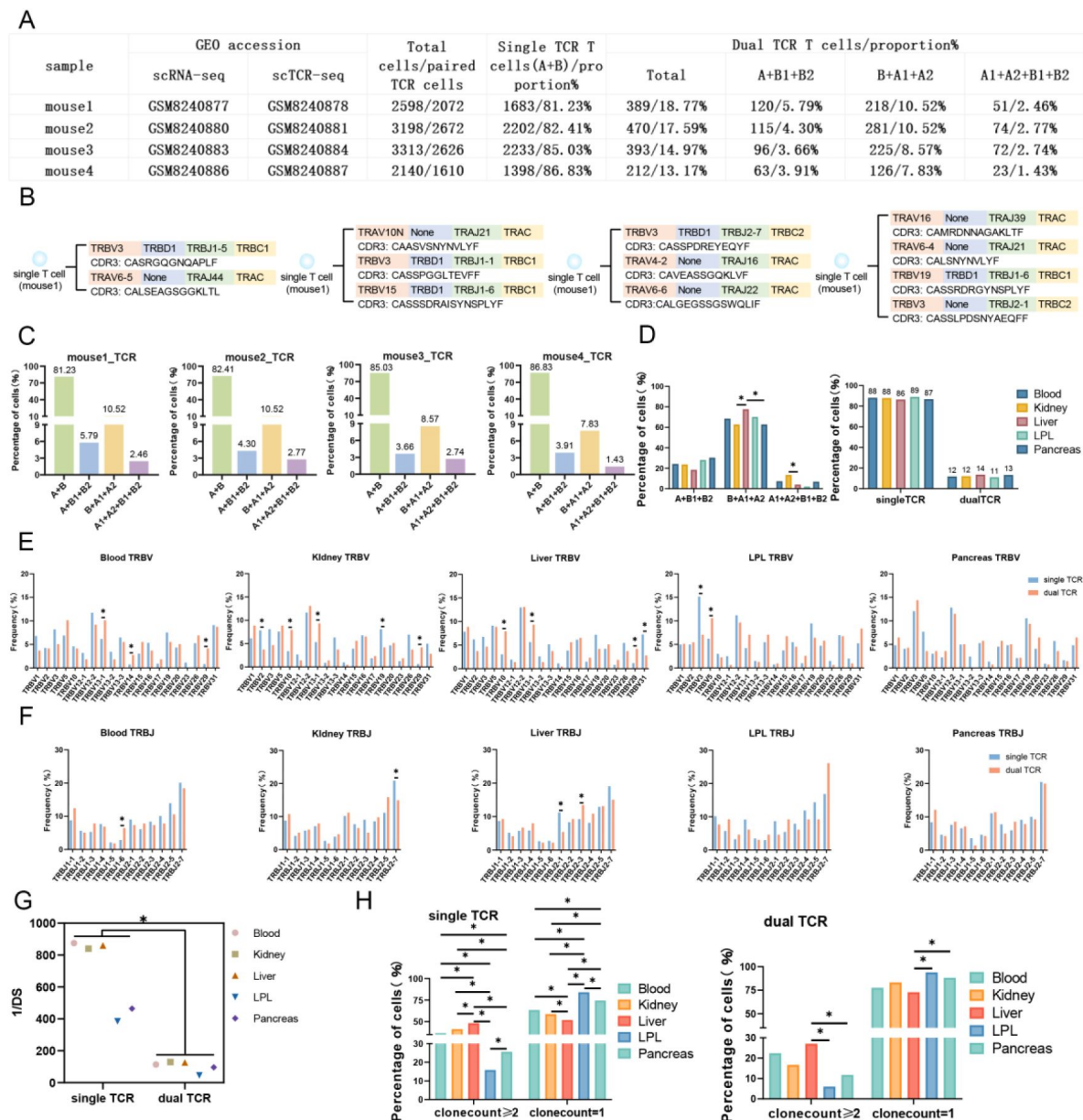


Figure 1.

Characterizes single Treg TCR and dual TCR Treg in the Blood, Kidney, Liver, LPL, and Pancreas of mice.

A, Mouse source names, GEO accession number, total number of paired TCR Treg cells, and number and proportion of TCR paired types. **B**, Four different TCR pairing types, VDJ gene family names and CDR3 AA sequences in single T cells. **C**, Proportion of TCR pairing types per mouse. **D**, Statistical analysis of dual TCR Treg cells and the relative proportion of single/dual TCR Treg Cells in Three TCR Pairing Types. **E**, Comparative analysis of single/dual TCR Treg cell V gene usage variation. **F**, Comparative analysis of single/dual TCR Treg cell J gene usage variation. **G**, Differential analysis of single/dual TCR Treg cell diversity among tissues. **H**, Differential analysis of single/dual TCR Treg cell clone expansion between tissues.

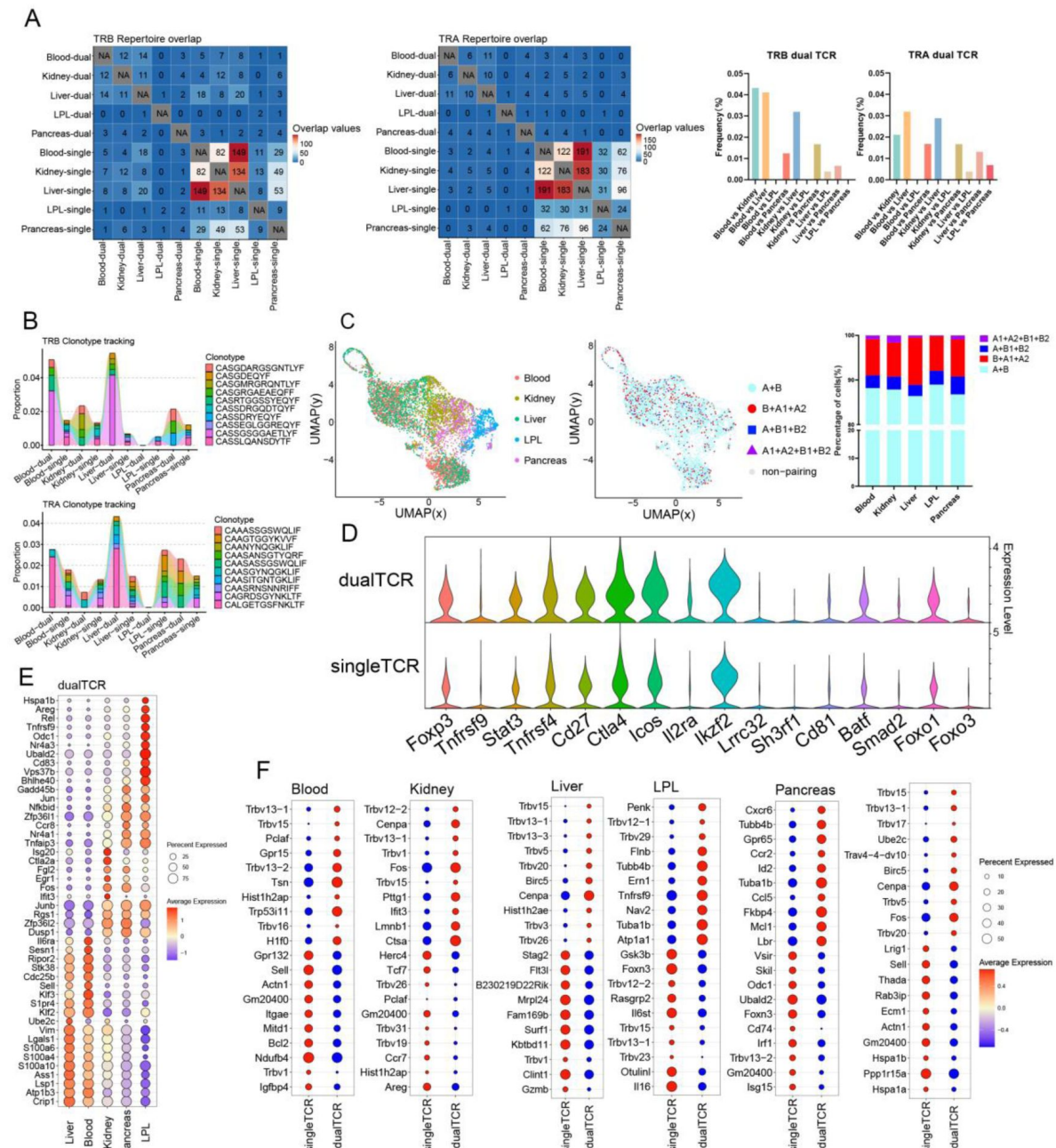


Figure 2.

Comparative analysis of single Treg TCR and dual TCR Treg CDR3 overlap and gene mRNA expression in mice Blood, Kidney, Liver, LPL, and Pancreas.

A, Clonal overlap analysis and jarcard index of TRA/B-chain CDR3 region in single /dual TCR Treg cells. **B**, Tracking results of the first 10 high-frequency overlapping sequences in the CDR3 region of the TRA/B chain of single /dual TCR Treg cells between tissues. **C**, Treg cell clustering results and proportion of Treg cells of the four TCR pairing types among tissues. **D**, Characterized mRNA expression in single/dual TCR Treg cells. **E**, Top 10 mRNA molecules highly expressed by dual TCR Treg cells in each tissue. **F**, Top 10 mRNA molecules highly expressed by single/dual TCR Treg cells in each tissue and overall.

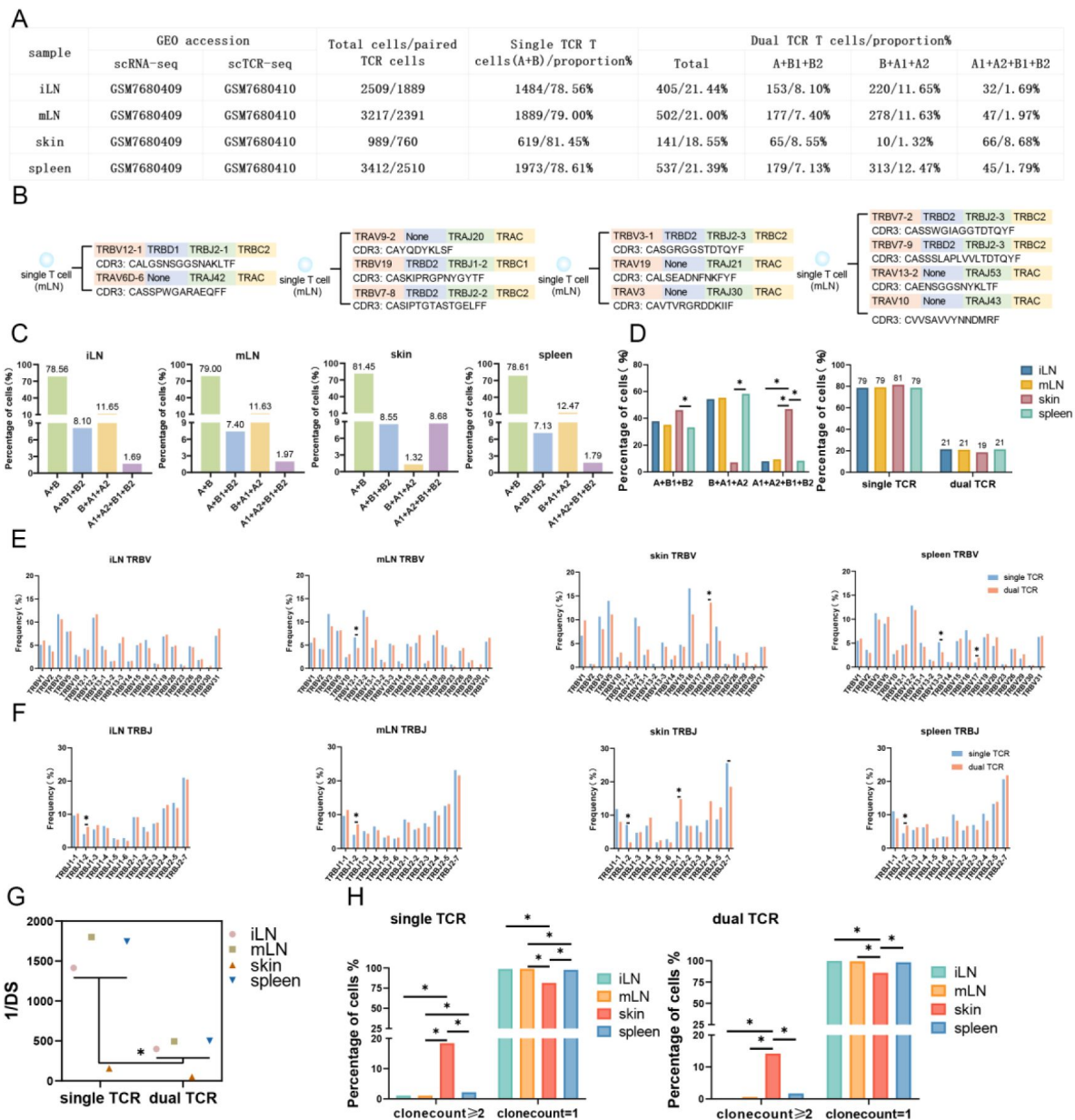


Figure 3.

Characterizes single Treg TCR and dual TCR Treg in the iLN, mLN, skin, spleen of mice.

A, Tissue source names, GEO accession number, total number of paired TCR T cells, and number and proportion of TCR paired types. **B**, Four different TCR pairing types, VDJ gene family names and CDR3 AA sequences in single T cells. **C**, Proportion of TCR pairing types per tissue. **D**, Statistical analysis of dual TCR T cells and the relative proportion of single/dual TCR T cells in three TCR pairing types. **E**, Comparative analysis of single/dual TCR T cell V gene usage variation. **F**, Comparative analysis of single/dual TCR T cell J gene usage variation. **G**, Differential analysis of single/dual TCR T cell diversity among tissues. **H**, Differential analysis of single/dual TCR T cell clone expansion between tissues.

2. Homogeneity and heterogeneity of TCR composition in mouse dual TCR Treg cells

VJ usage([Figure 1E](#) [↗](#), [Figure 3E](#) [↗](#)): The overall VJ family distribution pattern of single TCR Treg and dual TCR Treg between different tissues is basically consistent. Compared with single TCR Treg, dual TCR Treg displays specific preferences in V and J usage, mainly including: TRBV10 and TRBV13-1 in the kidney and liver, TRBV5 in LPL, TRBV19 in the skin; TRBJ2-3 in the liver, TRBJ2-1 in the skin, etc. showing a significant advantage in usage; TRBV13-1 in peripheral blood, kidney, and liver shows consistent advantage in usage; TRBJ1-2 in the spleen, iLN, and mLN shows consistent advantage in usage, etc.

Diversity of CDR3([Figure 1G](#) [↗](#), [Figure 3G](#) [↗](#)): The 1/DS of single TCR Treg CDR3 is significantly higher than that of dual TCR Treg; the CDR3 diversity of dual TCR Treg in lymphoid tissues (spleen, iLN, mLN) is higher than that in non-lymphoid tissues; the CDR3 diversity of dual TCR Treg in skin, LPL, and pancreas is very low.

Clonality of CDR3([Figure 1H](#) [↗](#), [Figure 3H](#) [↗](#)): The proportion of single TCR Treg with clonal proliferation greater than or equal to 2 is higher than that of dual TCR Treg; the proportion of dual TCR Treg with clonal proliferation greater than or equal to 2 in the liver and skin is significantly higher than that in other tissues.

Overlap of CDR3([Figure 2A/B](#) [↗](#), [Figure 4A/B](#) [↗](#)): A certain proportion of overlap exists between TRB CDR3 AA and TRA CDR3 AA in different tissues, with single TCR Treg significantly higher than dual TCR Treg, and a lower proportion of TRB CDR3 AA compared to TRA CDR3 AA. This suggests that TRA allelic inclusive recombination involves more CDR3 AA types in TCR pairing of Treg cells. Meanwhile, it was found that single and dual TCR Treg in different tissues can also share common CDR3 AA. This suggests that both single and dual TCR Treg can respond to specific antigen epitopes.

All three tissues, the liver, kidney, and pancreas, show a significantly high proportion of dual TCR Treg CDR3 AA overlap with peripheral blood, suggesting their origin from the influx of peripheral blood. iLN, mLN, and skin all show a significantly high proportion of dual TCR Treg CDR3 AA overlap with the spleen, suggesting their origin from the influx of the spleen; Intriguingly, there is also a high proportion of dual TCR Treg CDR3 AA overlap between the liver & kidney, kidney & pancreas, and iLN & mLN, suggesting that dual TCR Treg cells in these tissue sites may negatively regulate a shared set of antigens. Whereas the mechanism and significance of the high percentage of overlap exhibited between iLN and skin TRA CDR3 AA deserves further analysis.

3. Homogeneity of mouse dual TCR Treg cell phenotypes

Comparative analysis of Treg cells with complete scRNA-seq and scTCR-seq results showed that the TCR pairing types and proportion were consistent with the results of direct analysis of scTCR-seq; the proportion of dual TCR Treg in the skin was higher than that of dual TCR non-Treg ([Figure 2C](#) [↗](#), [Figure 4C](#) [↗](#)); in all tissues, the expression of signature molecules *Tnfrsf9*, *Stat3*, *Tnfrsf4*, *Cd27*, *Icos*, *Il2ra*, *Ikzf2*, *Lrrc32*, *Sh3rf1*, *Cd81*, *Baff*, *Smad2*, and *Foxo3* was identical between dual and single TCR Treg in all tissues, with slightly higher *Foxp3*, *Foxo1*, and *Ctla4* expression in dual TCR Treg than that in single TCR Treg([Figure 2D](#) [↗](#), [Figure 4D](#) [↗](#)).

4. Heterogeneity of mouse dual TCR Treg cell phenotypes

Compared with iLN, mLN, and spleen, more characteristic mRNA molecules were highly expressed in dual TCR Tregs in skin ([Figure 2E](#) [↗](#)), such as *Rora*, *Dusp1*, *Junb* involved in immune responses and inflammatory diseases; *S100A6* involved in the proliferation, differentiation, and migration of

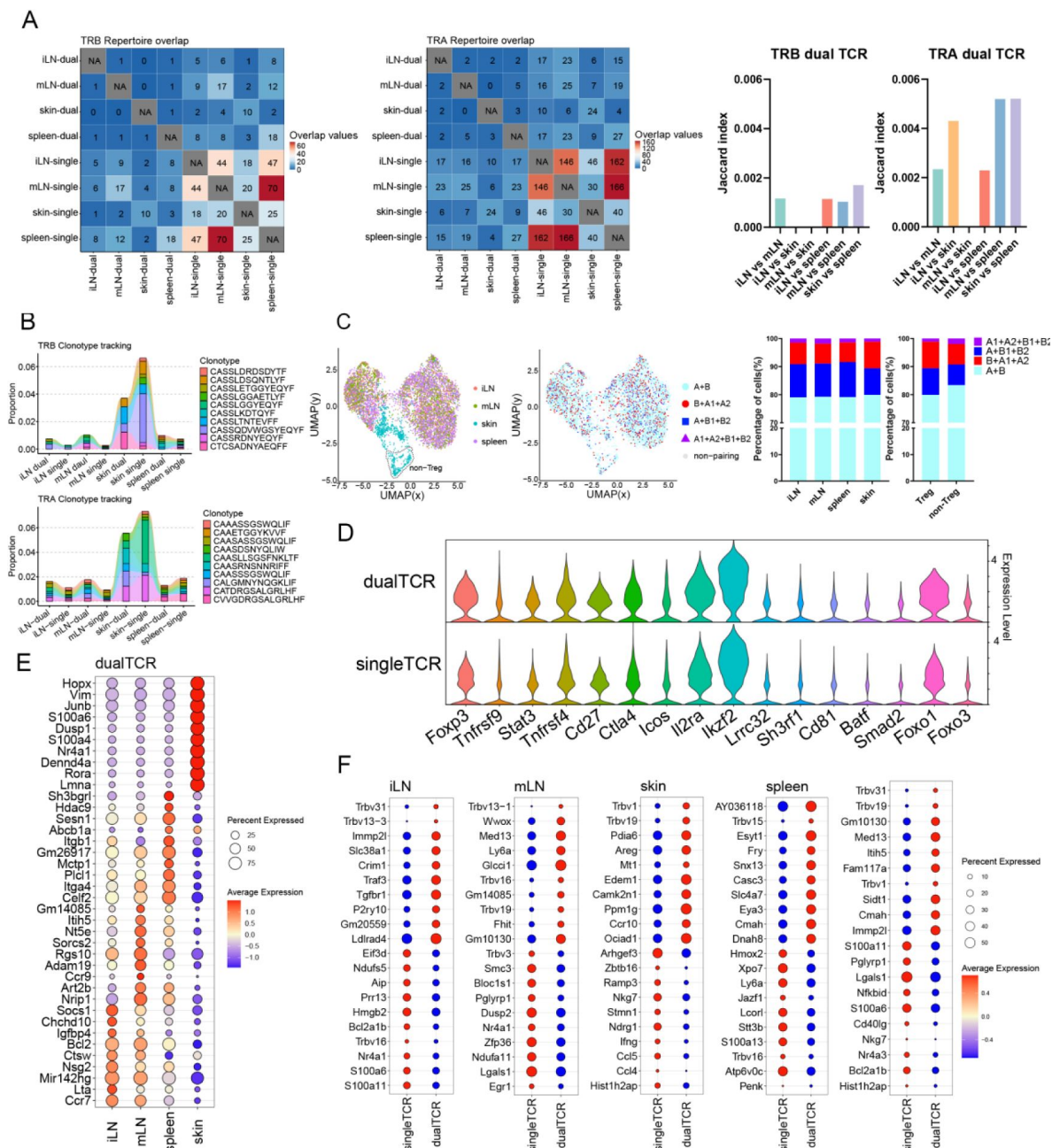


Figure 4.

Comparative analysis of single Treg TCR and dual TCR Treg CDR3 overlap and gene mRNA expression in mice iLN, mLN, skin, spleen.

A, Clonal overlap analysis and jaccard index of TRA/B-chain CDR3 region in single /dual TCR T cells. **B**, Tracking results of the first 10 high-frequency overlapping sequences in the CDR3 region of the TRA/B chain of single /dual TCR T cells between tissues. **C**, Treg cell clustering results and proportion of Treg cells of the four TCR pairing types among tissues. **D**, Characterized mRNA expression in single/dual TCR Treg cells. **E**, Top 10 mRNA molecules highly expressed by dual TCR Treg cells in each tissue. **F**, Top 10 mRNA molecules highly expressed by single/dual TCR Treg cells in each tissue and overall.

immune cells; *Hopx* and *Nr4a1* involved in the development of immune cells, etc., suggesting that dual TCR Treg in the skin may be involved in more specific response negative regulations.

Between Blood & Liver; Kidney & Pancreas (**Figure 2E** [↗](#)), iLN & mLN & spleen (**Figure 4E** [↗](#)), the mRNA expression of cytokines, cytokine receptors, and transcription factors in dual TCR Treg shows higher homogeneity, suggesting that dual TCR Treg in these tissue sites may play convergent negative immune response regulations.

Differential mRNA expression analysis between dual TCR Treg and single TCR Treg in each tissue (**Figure 2F** [↗](#), **Figure 4F** [↗](#)): showed significant high expressions of individual TRBV genes in different tissues, consistent with scTCR-seq results, suggesting specific dual TCR Treg lineage origins in different tissues; in addition to TRBV usage, each group's dual TCR Treg had significantly different characteristic molecule expressions from single TCR Treg, such as Blood: *H1f0* involved in immune cell proliferation and differentiation and *Hist1h2ap* related to immune cell activation were significantly increased; Kidney: *Fos* and *Cenpa* related to immune cell proliferation and differentiation and *Ifit3* involved in the body's antiviral response were significantly increased; Liver: *Cenpa* related to immune cell proliferation and *Birc5* involved in the body's anti-infection immune response were significantly increased; LPL: *Flnb* related to cell adhesion and migration regulation and *Atp1a1* related to cell proliferation, differentiation, and apoptosis regulation were significantly increased; Pancreas: *Ccl5*, *Ccr2*, *Cxcr6* related to immune cell migration and *Lbr* and *Id2* related to immune cell proliferation and differentiation, as well as *Fkbp4* involved in immune suppression signal transduction were significantly increased; iLN: *Tgfb1* involved in maintaining immune tolerance and regulating inflammatory responses and *Traf3* involved in immune cell activation and cytokine production were significantly increased; mLN: *Glcc1* involved in inflammation and immune cell migration was significantly increased; skin: *Areg* involved in immune cell proliferation, differentiation, migration, and immune suppression functions and *Ccr10* involved in T cell migration and positioning were significantly increased; spleen: *Casc3* related to T cell activation and proliferation was significantly increased. This suggests that dual TCR Treg and single TCR Treg in each specific tissue site may be involved in different negative immune response regulations.

Consistent high expression of molecular mRNA between different lymphoid tissues, such as: the presence of the *Nr4a1* gene related to T cell development, activation, and proliferation in both iLN and mLN tissues; partial sharing of molecular mRNA high expression between different non-lymphoid tissues, such as: the expression of *Trbv13-1* and *Trbv5* genes in Blood, Kidney, and Liver tissues, and the common expression of *Trbv1* in skin and Kidney, these results may suggest that dual TCR Treg cells in some tissues have specific lineage origins; and the common expression of the *Cenpa* gene in Kidney and Liver tissues may play a common immune effect; similarly, single TCR Treg also has similar gene expressions, such as the high expression of *Trbv1* in single TCR Treg cells in Blood and Liver tissues, and the high expression of *Hist1h2ap* related to immune cell activation in skin and Kidney.

Discussion

Currently, the proportion of dual TCR T cells is reported to be extremely variable across physiologic and pathologic states (0.1%-30%, etc.) due to differences in assay technology and methodology[4 [↗](#)][5 [↗](#)][6 [↗](#)][7 [↗](#)][8 [↗](#)][9 [↗](#)]. This study, for the first time in a large number of single Treg cell scRNA+TCR-seq studies, found that mouse lymphoid and non-lymphoid tissues both have a high proportion of dual TCR Treg cells, with lymphoid tissues being higher than non-lymphoid tissues, and the proportion of liver and pancreas dual TCR Treg cells in non-lymphoid tissues is relatively high. In addition, our analysis of dual TCR T cell pairing types revealed

significant differences between tissues. These findings not only suggest that lymphoid tissues have a high proportion of dual TCR Treg cells but also provide a comparative baseline and research direction for tissue-specific Treg cells in different locations.

The lower diversity of dual TCR Treg CDR3, the tissue preference in V/J gene usage, and the high clonal proliferation frequency in some tissues reflect the limited breadth of the dual TCR Treg subset response. These cells may play a complementary role in specific physiological or pathological negative regulation, and the tissue-specific preference for V/J usage suggests that dual TCR Treg cells may have a characteristic lineage similar to that of $\gamma\delta$ T cells. Current research indicates that $\gamma\delta$ T cells in different tissues show differences in VJ subfamily usage due to directed migration after specific VJ recombination at different times[14]. Dual TCR Treg cells may have certain lineage characteristics in V(D)J recombination and self-tolerance selection, and migrate to settle in different tissues. The higher clonal proliferation frequency of dual TCR Treg cells in the liver may exert a stronger inhibitory response to certain specific antigenic epitopes. The relatively higher immune tolerance of liver transplantation compared to other organ transplants may be related to the high proportion and clonal proliferation of dual TCR Treg cells in the liver.

The majority of single/dual TCR Treg cells exhibit consistent expression of characteristic genes, and the high expression of certain genes in dual TCR Treg cells (*Foxp3*, *Foxo1*, *Ctla4*) indicates that dual TCR Treg cells can function similarly to or more potently than single TCR Treg cells. In addition, specific or shared Trbv genes and mRNA molecule expression in dual TCR Treg cells has been identified in lymphoid or non-lymphoid tissues. Recently, Malte et al. used scATAC-seq technology to reveal organ-specific adaptation and conservation of tissue-resident immune cells such as Treg and Th17, and identify key transcription factors for multiple tissue-resident immune cells[15]. Overall, the homogeneity and heterogeneity of these characteristic molecules within Treg cells suggests their potential induction in response to different antigenic epitopes and corresponding negative regulatory functions.

This study, based on single-cell RNA sequencing and T cell receptor (scRNA+TCR-seq) data of Treg cells in lymphatic and non-lymphatic tissues of mice under physiological conditions, reveals for the first time the phenomenon of a large number of dual TCR Treg cells in mice. Moreover, the CDR3 regions of these dual TCR Treg cells exhibit certain tissue-specific or lineage-specific characteristics in terms of VJ usage, diversity, and clonality. Compared to single TCR Treg cells, dual TCR Treg cells highly express immune-inhibitory related molecules, such as *Foxo3*, *Foxo1*, *CD27*, *IL2RA*, and *Ikzf2*, etc. Dual TCR Treg cells in LPL and skin also have more mRNA differentially expressed characteristic molecules. These results provide a new Treg cell subset (dual TCR Treg) for studying phenotypic characteristics, TCR composition characteristics, etc. This study delves into the complex biological effects and mechanisms of Treg cells from the perspective of TCR mRNA expression, providing a new perspective for Treg cell research and also offering technical solutions for decoding the TCR pairing and characteristics of dual TCR T cells. However, this study still has some shortcomings, such as the lack of protein data for individual dual TCR Treg cells and research on human dual TCR Treg cells. Subsequent studies can use dual receptor transgenic reporter mice[16], combined with FCM, tetramers, and scCITE-seq technology, to provide a foundation for research at the TCR protein expression level. We hope that more laboratories will participate in the study of dual TCR Treg cells, especially to further clarify the diversity and specificity of the immune negative regulatory role of Treg cells in human lymphatic and non-lymphatic tissues.

Materials and methods

1. Research Subjects and Study Samples

(1) Blood, kidney, liver, LPL and pancreas tissues were collected from 4 C57BL/6 mice and CD4⁺Foxp3⁺ Thy1.1⁺ T reg cells were sorted for scRNA+TCR-seq analysis. (Source document: Oliver T et al. The tissue-resident regulatory T cell pool is shaped by transient multi-tissue migration and a conserved residency program. *Immunity*. 2024 Jul 9;57(7):1586-1602.e10. Shared Data Link: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE266111> ) (2) Spleen, iLN and mLN tissues from 4 mice were collected and sorted into CD4⁺CD25⁺Treg cells, while skin tissues from 4 mice were collected and sorted into CD3⁺ T cells (both Treg and non-Treg). These four tissue-sorted cells were analyzed by scRNA+TCR-seq. (Source document: Nedwed et al. Using combined single-cell gene expression, TCR sequencing and cell surface protein barcoding to characterize and track CD4⁺ T cell clones from murine tissues. *Front Immunol*. 2023 Oct 12;14:1241283. Shared Data Link: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE240041> )

2. The process of sharing raw data analysis

(1) Individual Treg cells (including CD3⁺ T cells in skin tissues) were screened for sequencing of both the TCR alpha chain (A) and TCR beta chain (B); (2) In a single T cell, two types of chains “can pair and assemble into single TCR” T cells named “A+B”; Three or more types of chains “can pair and assemble into dual TCR” T cells named “A+B1+B2; B+A1+A2; A1+A2+B1+B2”, (3) The proportions of total single TCR Treg and dual TCR Treg were calculated for each mouse or each tissue.

3. Examples of single T-cell TCR pairing types

From the two data sets, single T cell was selected to display four different TCR pairing types, VDJ gene family names and CDR3 AA sequences: A+B; A+B1+B2; B+A1+A2; and A1+A2+B1+B2.

4. Analysis of TCR pairing types and V-J usage

Statistically analyze the proportions of five TCR T cell types in each sample, and use SPSS software to perform statistical analysis on the proportions of the three dual TCR T cell types and the single/dual TCR T cell ratios according to tissue origin; classify and calculate the proportions of all corresponding sequences in the ‘β-V’ and ‘β-J’ columns of the single and dual TCR T cell tables, and use SPSS software to conduct differential VJ usage analysis between single/dual TCR T cells.

5. CDR3 diversity and clonality analysis

Use the inverse Simpson’s index to assess TCR Treg cell diversity and perform statistical analysis; define clonal expansion as clone count ≥ 2, and analyze the proportion of clonal expansion of single/dual TCR Treg cells in the two data sets.

6. CDR3 overlap analysis

The TRA CDR3 AA and TRB CDR3 AA of single TCR Treg cells and dual TCR Treg cells from different tissue sources were compared for overlap using the immunarch R package, and the Jacard index was used to analyze the frequency of overlap of CDR3 AA between each tissues pair; the first 10 high-frequency overlapping TRB CDR3 AA and TRA CDR3 AA sequences were traced and analyzed between single TCR Treg cells and dual TCR Treg cells from different tissue sources.

7. Combined scRNA+TCR-seq analysis

The scRNA-seq results of the two datasets were quality-controlled to exclude data with nFeature > 5,000, nCount < 3,000 and > 20,000, and percent.MT > 5 (skin, spleen, iLN and mLN tissue datasets) or 10 (blood, kidney, liver, LPL and pancreas tissues). Subsequently, the “DoubletFinder” R package was used to remove doublets from four tissue sources (including skin), and doublets/negatives were eliminated from five tissue sources (including LPL) based on HTO labels. T cells were then clustered and analyzed from different tissue sources after quality control. The scTCR-seq results

were analyzed in conjunction with the scRNA-seq results, and only Treg cells meeting both criteria were included. Cells containing scTCR-seq and scRNA-seq data were displayed in UMAP plots and analyzed for the proportion of single TCR Treg cells and dual TCR Treg cells from different tissue sources. In this analysis, skin tissue was divided into Treg and non-Treg for comparison.

8. Cytokine, cytokine receptor, and transcription factor expression analysis

Comparative analysis of the homogeneity of Treg cell signature molecules expressed in single TCR Treg and dual TCR Treg; Comparative analysis of the heterogeneity of the expression of the top 10 mRNA molecules of dual TCR Treg in different tissue origins; Comparative analysis of the heterogeneity of the expression of single TCR Treg and dual TCR Treg top 10 mRNA molecules in each tissue; Comparative analysis of the heterogeneity of the expression of total single TCR Treg and total dual TCR Treg top 10 mRNA molecules.

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Additional information

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Authorship Contributions

Xinsheng Yao completed the experimental design and paper writing, Yuanyuanxu, Qipeng, Xiaoping Lu, completed the data analysis and chart making.

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Reviewer #1 (Public review):

Summary:

This study presents findings on dual TCR regulatory T cells (Tregs) using previously published single-cell RNA and TCR sequencing datasets. The authors aimed to quantify dual TCR Tregs in different tissues and analyze their characteristics. Rather than perform the difficult experiments needed to ascertain the functional role of dual receptors, this study relies entirely on scRNA-VDJ-seq data published by two other groups. The findings primarily confirm prior work rather than provide new insights, and the methodology has significant weaknesses that limit the study's impact. We have concerns about the scientific integrity of this work.

Strengths:

- (1) The use of single-cell RNA and TCR sequencing is appropriate for addressing potential relationships between gene expression and dual TCR.
- (2) The data confirm the presence of dual TCR Tregs in various tissues, with proportions ranging from 10.1% to 21.4%, aligning with earlier observations in $\alpha\beta$ T cells.
- (3) Tissue-specific patterns of TCR gene usage are reported, which could be of interest to researchers studying T cell adaptation, although these were more rigorously analyzed in the original works.

Weaknesses

- (1) Lack of Novelty: The primary findings do not substantially advance our understanding of dual TCR expression, as similar results have been reported previously in other contexts.
- (2) Incomplete Evidence: The claims about tissue-specific differences lack sufficient controls (e.g., comparison with conventional T cells) and functional validation (e.g., cell surface expression of dual TCRs).
- (3) Methodological Weaknesses: The diversity analysis does not account for sample size differences, and the clonal analysis conflates counts and clonotypes, leading to potential misinterpretation.
- (4) Insufficient Transparency: The sequence analysis pipeline is inadequately described, and the study lacks reproducibility features such as shared code and data.
- (5) Weak Gene Expression Analysis: No statistical validation is provided for differential gene expression, and the UMAP plots fail to reveal meaningful clustering patterns.
- (6) A quick online search reveals that the same authors have repeated their approach of reanalysing other scientists' publicly available scRNA-VDJ-seq data in six other publications:

- (1) Peng, Q., Xu, Y. & Yao, X. scRNA+ TCR-seq revealed dual TCR T cells antitumor response in the TME of NSCLC. *J Immunother Cancer* 12 (2024). <https://doi.org/10.1136/jitc-2024-009376>
- (2) Wang, H., Li, J., Xu, Y. & Yao, X. scRNA + BCR-seq identifies proportions and characteristics of dual BCR B cells in the peritoneal cavity of mice and peripheral blood of healthy human donors across different ages. *Immun Ageing* 21, 90 (2024). <https://doi.org/10.1186/s12979-024-00493-6>
- (3) Xu, Y. et al. scRNA+TCR-seq reveals the pivotal role of dual receptor T lymphocytes in the pathogenesis of Kawasaki disease and during IVIG treatment. *Front Immunol* 15, 1457687 (2024). <https://doi.org/10.3389/fimmu.2024.1457687>
- (4) Yuanyuanxu, Qipeng, Qingqingma & Yao, X. scRNA + TCR-seq revealed the dual TCR pTh17 and Treg T cells involvement in autoimmune response in ankylosing spondylitis. *Int Immunopharmacol* 135, 112279 (2024). <https://doi.org/10.1016/j.intimp.2024.112279>
- (5) Zhu, L. et al. scRNA-seq revealed the special TCR beta & alpha V(D)J allelic inclusion rearrangement and the high proportion dual (or more) TCR-expressing cells. *Cell Death Dis* 14, 487 (2023). <https://doi.org/10.1038/s41419-023-06004-7>
- (6) Zhu, L., Peng, Q., Wu, Y. & Yao, X. scBCR-seq revealed a special and novel IG H&L V(D)J allelic inclusion rearrangement and the high proportion dual BCR expressing B cells. *Cell Mol Life Sci* 80, 319 (2023). <https://doi.org/10.1007/s00018-023-04973-8>

In other words, the approach used here seems to be focused on quick re-analyses of publicly available data without further validation and/or exploration

Appraisal of the Study's Aims and Conclusions:

The authors set out to analyze dual TCR Tregs across tissues, but the lack of robust controls, incomplete analyses, and insufficient novelty limit the study's ability to achieve its aims. The results confirm prior findings but do not provide compelling evidence to support the broader claims about the characteristics or significance of dual TCR Tregs.

Impact and Utility:

While the study provides a descriptive analysis of dual TCR Tregs, its limited novelty and methodological weaknesses reduce its likely impact on the field. The methods and data could have utility for researchers interested in tissue-specific TCR gene usage, but additional rigor is required to make the findings broadly applicable.

<https://doi.org/10.7554/eLife.105504.1.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript, "scRNA+TCR-seq Reveals the Proportion and Characteristics of Dual TCR Treg Cells in Mouse Lymphoid and Non-lymphoid Tissues" by Xu and Peng, et al. investigates whether co-expression of 2 T cell receptor (TCR) clonotypes can be detected in FoxP3+ regulatory CD4+ T cells (Tregs) and if it is associated with identifiable phenotypic effects. This paper presents data reanalyzing publicly available single-cell TCR sequencing and transcriptional analysis, convincingly demonstrating that dual TCR co-expression can be detected in Tregs, both in peripheral circulation as well as among Tregs in tissues. They then compare metrics of TCR diversity between single-TCR and dual TCR Tregs, as well as between Tregs in different anatomic compartments, finding the TCR repertoires to be generally similar

though with dual TCR Tregs exhibiting a less diverse repertoire and some moderate differences in clonal expansion in different anatomic compartments. Finally, they examine the transcriptional profile of dual TCR Tregs in these datasets, finding some potential differences in the expression of key Treg genes such as *Foxp3*, *CTLA4*, *Foxo3*, *Foxo1*, *CD27*, *IL2RA*, and *Ikzf2* associated with dual TCR-expressing Tregs, which the authors postulate implies a potential functional benefit for dual TCR expression in Tregs.

Strengths:

This report examines an interesting and potentially biologically significant question, given recent demonstrations that dual TCR co-expression is a much more common phenomenon than previously appreciated (approximately 15-20% of T cells) and that dual TCR co-expression has been associated with significant effects on the thymic development and antigenic reactivity of T cells. This investigation leverages large existing datasets of single-cell TCRseq/RNAseq to address dual TCR expression in Tregs. The identification and characterization of dual TCR Tregs is rigorously demonstrated and presented, providing convincing new evidence of their existence.

Weaknesses:

The existence of dual TCR expression by Tregs has previously been demonstrated in mice and humans (Reference #18 and Tuovinen. 2006. *Blood*. 108:4063; Schuldt. 2017. *J Immunol*. 199:33, both omitted from references). The presented results should be considered in the context of these prior important findings.

This demonstration of dual TCR Tregs is notable, though the authors do not compare the frequency of dual TCR co-expression by Tregs with non-Tregs. This limits interpreting the findings in the context of what is known about dual TCR co-expression in T cells.

Comparison of gene expression by single- and dual TCR Tregs is of interest, but as presented is difficult to interpret. Statistical analyses need to be performed to provide statistical confidence that the observed differences are true.

The interpretations of the gene expression analyses are somewhat simplistic, focusing on the single-gene expression of some genes known to have a function in Tregs. However, the investigators miss an opportunity to examine larger patterns of coordinated gene expression associated with developmental pathways and differential function in Tregs (Yang. 2015. *Science*. 348:589; Li. 2016. *Nat Rev Immunol*. Wyss. 2016. 16:220; *Nat Immunol*. 17:1093; Zenmour. 2018. *Nat Immunol*. 19:291).

<https://doi.org/10.7554/eLife.105504.1.sa1>

Reviewer #3 (Public review):

Summary:

This study addressed the TCR pairing types and CDR3 characteristics of Treg cells. By analyzing scRNA and TCR-seq data, it claims that 10-20% of dual TCR Treg cells exist in mouse lymphoid and non-lymphoid tissues and suggests that dual TCR Treg cells in different tissues may play complex biological functions.

Strengths:

The study addresses an interesting question of how dual-TCR-expressing Treg cells play roles in tissues.

Weaknesses:

This study is inadequate, particularly regarding data interpretation, statistical rigor, and the discussion of the functional significance of Dual TCR Tregs.

Major Comments:

(1) Definition of Dual TCR and Validity of Doublet Removal

This study analyzes Treg cells with Dual TCR, but it is not clearly stated how the possibility of doublet cells was eliminated. The authors mention using DoubletFinder for detecting doublets in scRNA-seq data, but is this method alone sufficient?

We strongly recommend reporting the details of doublet removal and data quality assessment in the Supplementary Data.

(2) Inconsistency in the Proportion of Dual TCR T Cells in the Skin Across Figures

In Figure 3D, the proportion of Dual TCR T cells (A1+A2+B1+B2) in the skin is reported to be very high compared to other tissues. However, in Figure 4C, the proportion appears lower than in other tissues, which may be due to contamination by non-Tregs. The authors should clarify why it was necessary to include non-Tregs as a target for analysis in this study. Additionally, the sensitivity of scRNA-seq and TCR-seq may vary between tissues and may also be affected by RNA quality and sequencing depth in skin samples, so the impact of measurement bias should be assessed.

(3) Issue of Cell Contamination

In Figure 2A, the data suggest a high overlap between blood, kidney, and liver samples, likely due to contamination. Can the authors effectively remove this effect? If the dataset allows, distinguishing between blood-derived and tissue-resident Tregs would significantly enhance the reliability of the findings. Otherwise, it would be difficult to separate biological signals from contamination noise, making interpretation challenging.

(4) Inconsistency Between CDR3 Overlap and TCR Diversity

The manuscript states that Single TCR Tregs have a higher CDR3 overlap, but this contradicts the reported data that Dual TCR Tregs exhibit lower TCR diversity (higher 1/DS score). Typically, when TCR diversity is low (i.e., specific clones are concentrated), CDR3 overlap is expected to increase. The authors should carefully address this discrepancy and discuss possible explanations.

(5) Functional Evaluation of Dual TCR Tregs

This study indicates gene expression differences among tissue-resident Dual TCR T cells, but there is no experimental validation of their functional significance. Including functional assays, such as suppression assays or cytokine secretion analysis, would greatly enhance the study's impact.

(6) Appropriateness of Statistical Analysis

When discussing increases or decreases in gene expression and cell proportions (e.g., Figure 2D), the statistical methods used (e.g., t-test, Wilcoxon, FDR correction) should be explicitly described. They should provide detailed information on the statistical tests applied to each analysis.

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